FlowTrack

Project Management System

Version: 1.0.0
Type: Full-Stack Web Application
Stack: MongoDB · Express · React· Node.js · Docker
Date: March 29, 2026

1. Project Description & Introduction

1.1 Overview

FlowTrack is a full-stack project management web application built to streamline team collaboration, task tracking, and project delivery. Inspired by tools like Jira and Trello, FlowTrack is designed specifically for small to medium-sized software development teams that need a lightweight yet powerful system for managing projects, assigning tasks, and monitoring progress in real time.

The system implements a role-based access control (RBAC) model with three distinct roles — Admin, Team Lead, and User — each with clearly defined permissions that prevent conflicts of interest and ensure data integrity throughout the workflow.

1.2 Technical Architecture

FlowTrack follows a modern client-server architecture with a decoupled frontend and backend, containerized using Docker for consistent deployment across environments.

Feature	Description	Access
Frontend	React 19 with Vite, React Router, Axios, Tailwind CSS, React Toastify	All roles
Backend	Node.js with Express.js REST API, cookie-based JWT authentication	Server-side
Database	MongoDB with Mongoose ODM for schema modeling and validation	Server-side
Auth	httpOnly cookie-based JWT — secure, XSS-resistant session management	All roles
Deployment	Containerized deployment using Docker Compose	DevOps

1.3 Problem Statement

Many small development teams struggle with:

- Lack of visibility into task status and project progress
- No formal approval process for completed work
- Over-privileged team members creating conflicts of interest
- No centralized system for team and project organization

FlowTrack addresses all of these problems through structured workflows, role separation, and secure authentication.

2. Project Scope & Key Features

2.1 Scope

FlowTrack covers the full lifecycle of project and task management within a team, from team creation through to task completion and approval. The application is scoped for internal organizational use and is not designed as a public-facing SaaS product.

In Scope

- User registration, authentication, and session management
- Role-based access control (Admin, Team Lead, User)
- Team creation and member management
- Project creation with team and member assignment
- Task creation, assignment, status tracking, and notes
- Task submission and admin approval/rejection workflow
- Admin panel for user management and role assignment
- Dockerized deployment with nginx reverse proxy
- httpOnly cookie-based JWT authentication

Out of Scope

- Real-time notifications (WebSockets/push notifications)
- File or attachment uploads on tasks
- Time tracking and reporting dashboards
- Mobile native application
- Multi-tenant / SaaS architecture

2.2 Role-Based Access Control

The system enforces strict role separation to prevent conflicts of interest and ensure proper oversight:

Role	Permissions	Restrictions
Admin	Create teams, projects, tasks. Manage users. Approve/reject task submissions. Full CRUD on all entities.	None — full system access

Role	Permissions	Restrictions
Team Lead	View team projects and tasks. Add notes to tasks. Can add/remove a member from their team.	Cannot create, edit, delete, approve, or reject anything
User	View assigned tasks. Update task status. Add notes. Submit tasks for approval.	Read-only on projects/teams. Cannot manage any entities

2.3 Key Features

Authentication & Security

- httpOnly cookie-based JWT — tokens are inaccessible to JavaScript, preventing XSS attacks
- 1-hour token expiry with automatic logout on token failure
- Session restored on page reload via /auth/me endpoint
- Logout clears cookie server-side — no token persists after logout

Team Management

- Admin can create teams with a unique name (enforced at DB level)
- Each team has one designated Team Lead and multiple members
- A person can be Team Lead of one team and a regular member of another simultaneously
- Admin can add/remove members from teams

Project Management

- Projects are linked to a team and can have multiple assigned members
- Each project has a status (Planning, In Progress, Completed, On Hold) and priority level
- GitHub repository link can be attached to a project
- Members are selected from the project's team only
- Only admins can create, edit, or delete projects

Task Management & Approval Workflow

- Tasks are assigned to specific project members by the admin
- Tasks progress through statuses: Pending → In Progress → Review → Completed or Bug
- Users submit tasks for approval via a 'Submit for Approval' button

- Admin reviews submitted tasks and can Approve (marks complete) or Reject (with mandatory reason)
- Rejected tasks return to In Progress with the rejection reason displayed prominently
- Notes thread on each task captures the full audit trail of status changes and comments
- Tags can be added to tasks for categorisation

Admin Panel

- View all registered users with join date and role
- Promote users to Admin or demote to User
- Delete non-admin users from the system
- Admin accounts are protected from deletion

3. Workflow Diagrams

3.1 User Authentication Flow

The following describes the complete authentication lifecycle:

Authentication Flow

1. User visits the app → AuthContext calls GET /auth/me

2a. Valid cookie exists → server returns user data → user proceeds to dashboard

2b. No cookie / expired → server returns 401 → user redirected to /login

3. User submits login form → POST /auth/login
4. Server validates credentials → sets httpOnly cookie → returns user object
5. AuthContext stores user in React state → app re-renders with user context
6. All subsequent API requests send cookie automatically (withCredentials: true)
7. On logout → POST /auth/logout → server clears cookie → state cleared → redirect /login
8. On token expiry → any API call returns 401 → interceptor redirects to /login

3.2 Task Lifecycle Workflow

ADMIN	TEAM LEAD	USER	SYSTEM
Creates task and assigns to user	Views task in their team	Receives task as Pending	Status: pending
	Can view task details	Updates to In Progress	Status: in_progress
	Works on task, adds notes	Works on task, adds notes	Notes recorded with timestamp
	Clicks Submit for Approval	Clicks Submit for Approval	Status: review
Sees task in Review state	Can see task is in review	Sees Awaiting Approval banner	
Approves or Rejects task			
Approve → task complete	Sees Completed status	Sees Completed status	Status: completed
Reject → adds reason		Sees rejection reason banner	Status: in_progress

		Fixes and resubmits	Status: review again

3.3 Team & Project Setup Flow

The admin follows this sequence to set up a team and assign work:

1. Admin creates a Team — assigns a Team Lead and initial members
2. Admin creates a Project — links it to a team and selects members from that team
3. Admin creates Tasks — assigns each to a specific project member with a deadline and priority
4. Users work on tasks and update their status
5. Users submit completed tasks for admin review
6. Admin approves or rejects — rejected tasks loop back to step 4

4. Conclusion & Summary of Requirements

4.1 Summary

FlowTrack successfully delivers a structured, secure, and role-aware project management platform. The system enforces a clear hierarchy of permissions that prevents common organizational problems such as self-approval of work, unauthorized access to management functions, and insecure session handling.

The application has been built with a production-ready mindset from the ground up, using httpOnly cookies for security, Docker for consistent deployment, nginx as a reverse proxy, and MongoDB for flexible data modeling. Every design decision has been made with both developer experience and end-user security in mind.

4.2 Functional Requirements Summary

#	Requirement	Status
01	User registration with email and password	Implemented
02	JWT authentication via httpOnly cookies	Implemented
03	Session persistence via /auth/me on page reload	Implemented
04	Role-based access control (Admin / Team Lead / User)	Implemented
05	Team Lead status derived dynamically from team membership	Implemented
06	Admin can create, update, delete teams	Implemented
07	Unique team name enforcement at database level	Implemented
08	Admin can create projects linked to teams	Implemented
09	Project members selected from team members only	Implemented
10	Admin can assign tasks to project members	Implemented
11	Task status lifecycle: Pending → In Progress → Review → Completed	Implemented
12	User submits task for approval (Submit for Approval button)	Implemented
13	Admin approves task → status set to Completed	Implemented
14	Admin rejects task with mandatory reason → status reset to In Progress	Implemented

15	Rejection reason displayed prominently to assignee	Implemented
16	Notes thread on tasks with timestamps and author	Implemented
17	Admin panel for user role management and deletion	Implemented
18	Admin accounts protected from deletion	Implemented
19	Docker Compose deployment	Implemented

4.3 Non-Functional Requirements

- Security — httpOnly cookies, role-based middleware on all protected routes, no sensitive data in responses
- Scalability — stateless REST API, MongoDB horizontal scaling support, Docker containerization
- Maintainability — modular MVC architecture, separate route/controller/model layers
- Usability — responsive UI, clear error messages, toast notifications for all actions
- Deployability — single docker-compose up command to run the full stack

5. Future Enhancements

The following features have been identified as high-value additions for future versions of FlowTrack:

5.1 Real-Time Notifications

Integrate WebSockets (Socket.io) to send live notifications to users when tasks are assigned, submitted, approved, or rejected. This eliminates the need for manual page refreshes and significantly improves team responsiveness.

5.2 File Attachments on Tasks

Allow users to upload files (screenshots, documents, code snippets) directly to tasks. This would leverage cloud storage (AWS S3 or Cloudinary) and provide a centralized location for all task-related assets.

5.3 Analytics & Reporting Dashboard

Build a comprehensive analytics dashboard for admins showing team velocity, task completion rates, overdue task trends, and individual contributor performance. This would use chart libraries like Recharts or Chart.js to visualize MongoDB aggregation pipeline results.

5.4 GitHub Integration

Connect tasks to GitHub pull requests and commits via webhooks. Automatically update task status when a linked PR is merged, and show commit history directly on the task detail page. This would deeply integrate FlowTrack into the development workflow.

5.5 Email Notifications

Send automated email notifications using Nodemailer or SendGrid for key events: task assignment, approaching deadlines, task rejection, and approval. This ensures team members stay informed even when not actively using the application.

5.6 Time Tracking

Add the ability for users to log time spent on tasks. Admins and team leads would be able to view time reports per project and per user, enabling better workload balancing and project estimation for future sprints.

5.8 Two-Factor Authentication (2FA)

Enhance security with optional TOTP-based 2FA using authenticator apps like Google Authenticator. This is especially important for admin accounts which have full system access.

5.9 Audit Log

Maintain a complete audit trail of all significant actions in the system (user created, role changed, task approved, project deleted, etc.) accessible to admins. This improves accountability and aids in debugging and compliance.

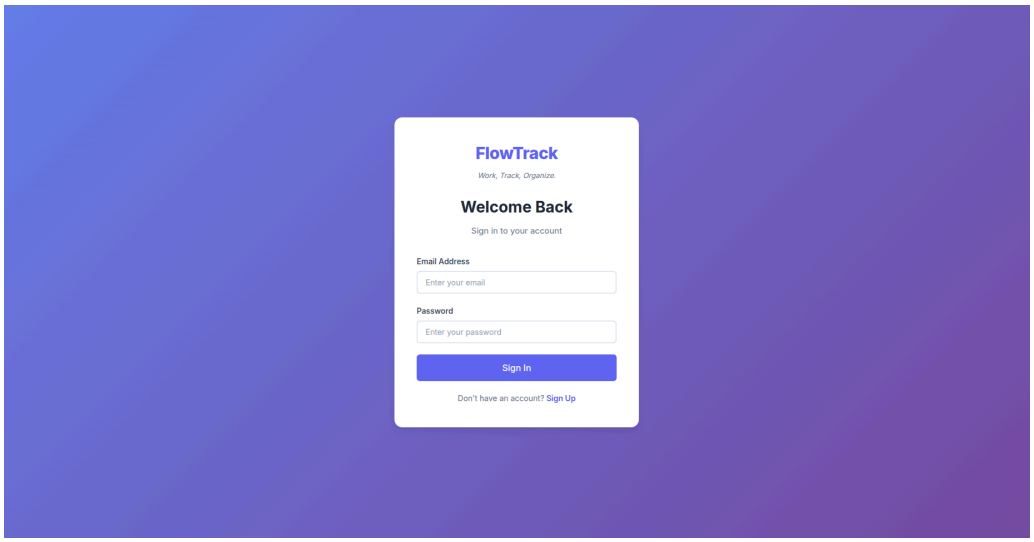
5.10 Multi-Language Support (i18n)

Internationalise the frontend using react-i18next to support multiple languages. This would make FlowTrack accessible to non-English speaking teams and broaden its potential user base.

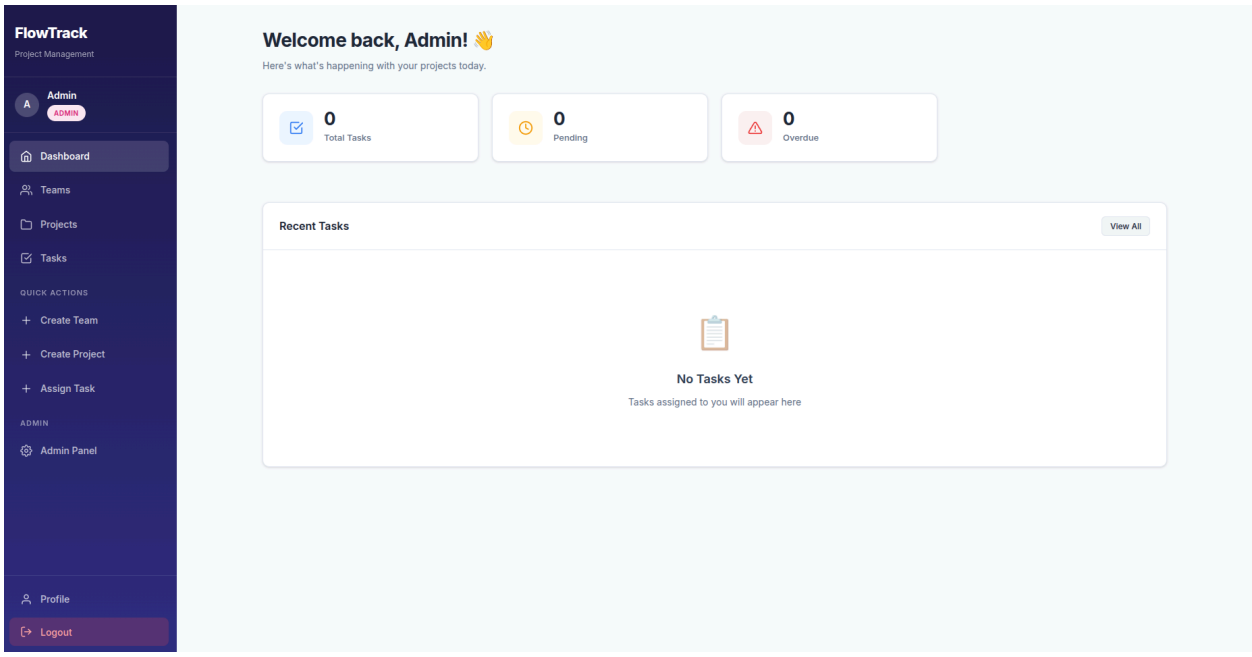
FlowTrack — Work,Track,Analyze.

The below given are the screenshots of the application.

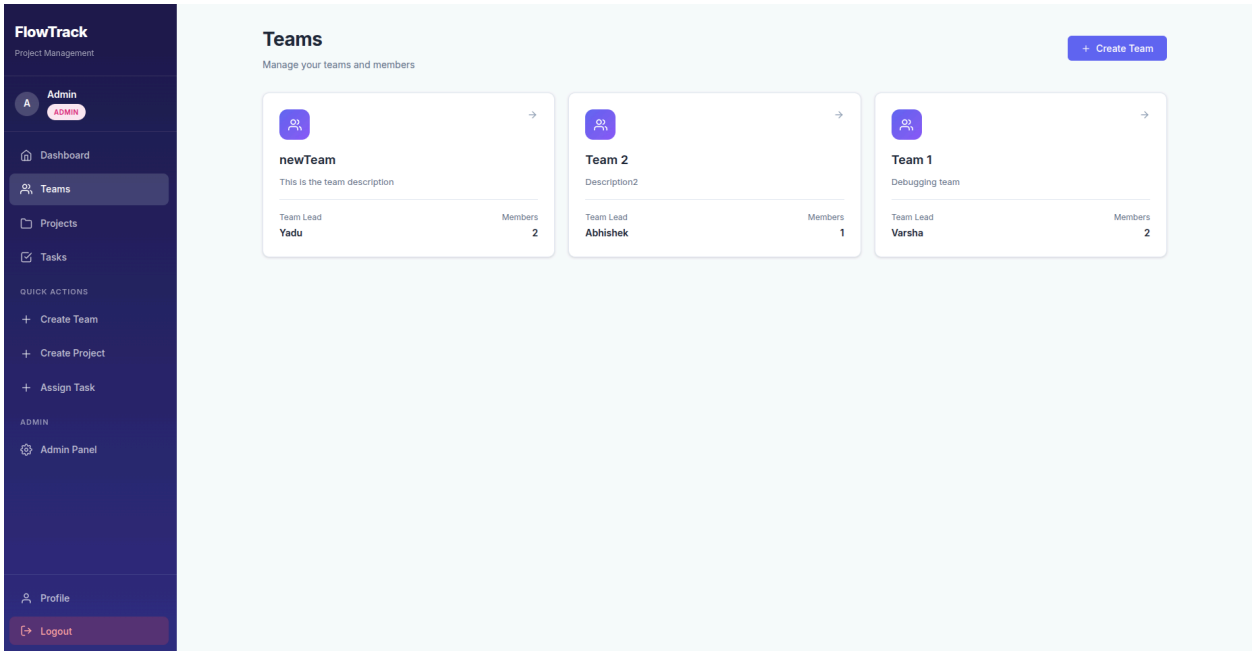
Login



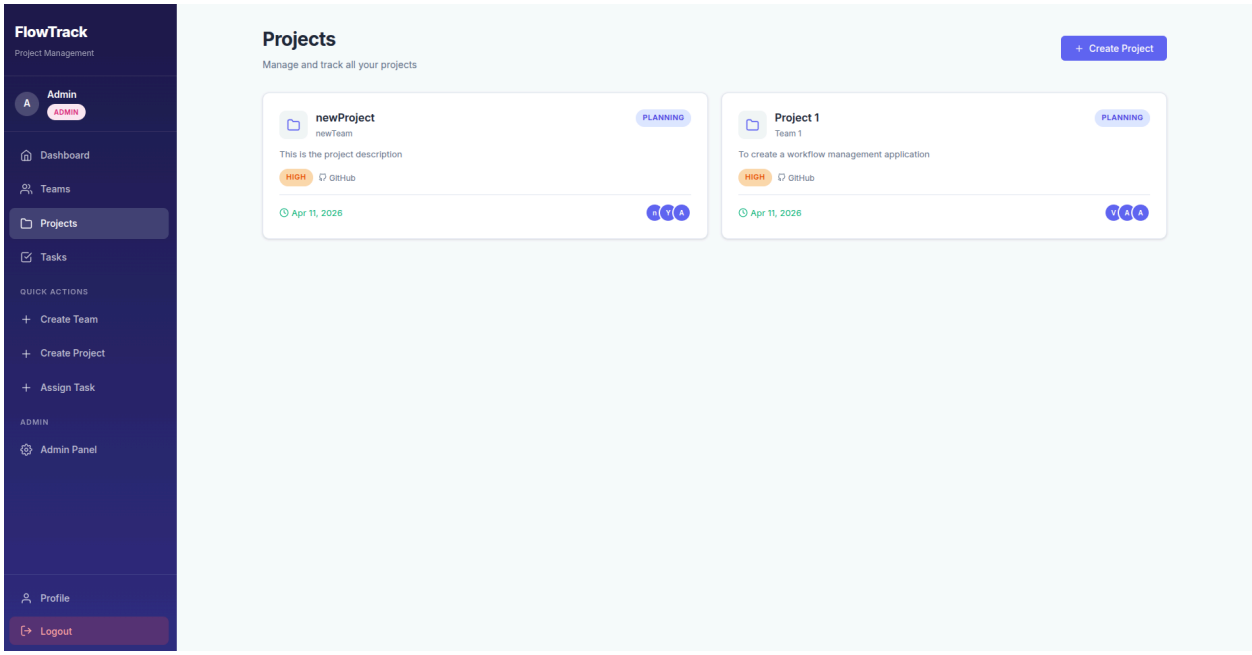
Dashboard



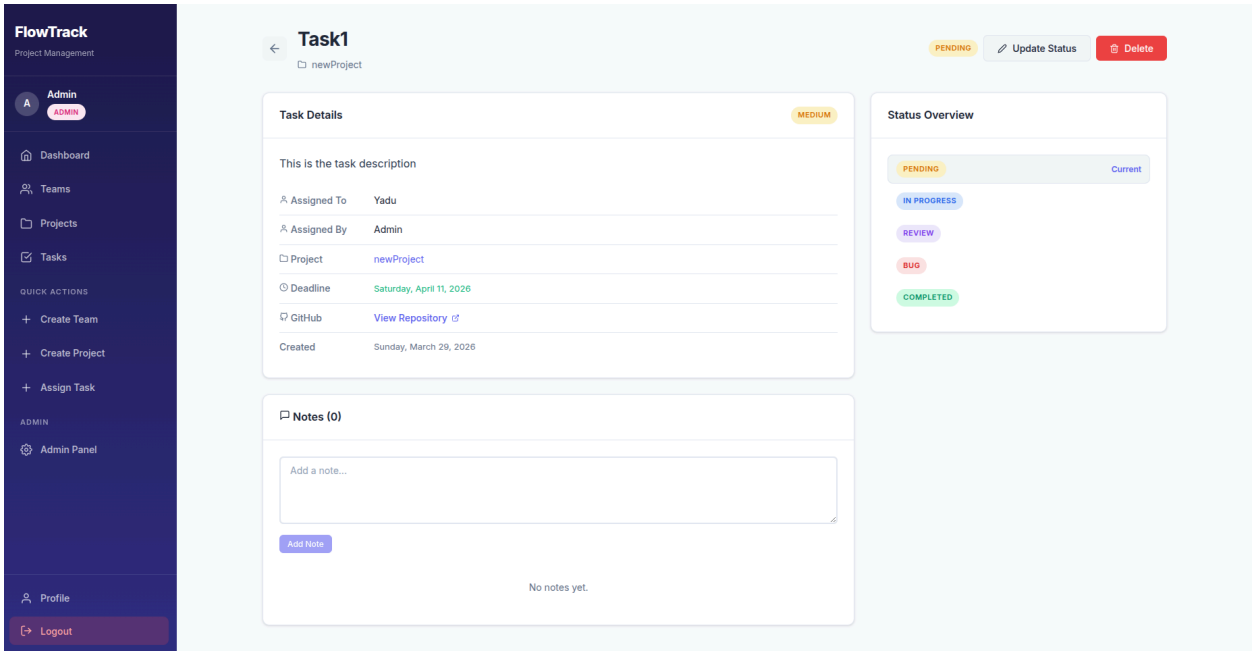
Teams



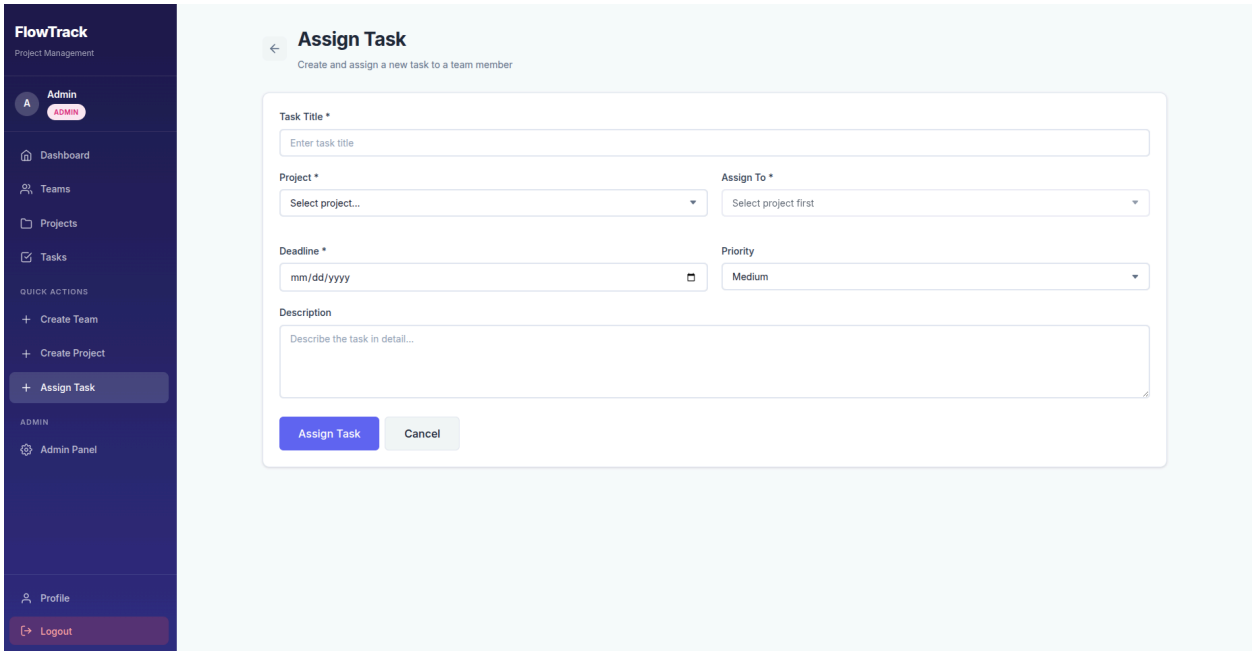
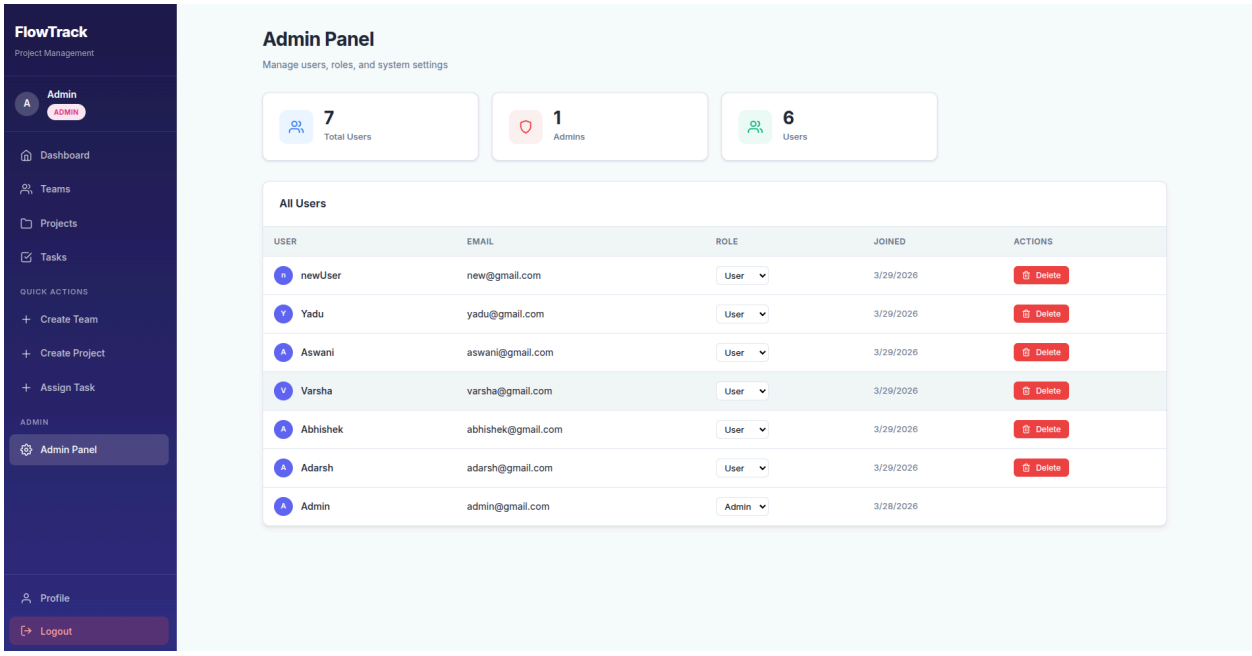
Projects



Task Details



Assign Tasks



Admin Pannel